```
import sys
%cd /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver
!{sys.executable} -u src/main_liver.py | tee liver_train_v2.log
```

```
/nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver
/nlsasfs/home/aikosh/prod-aikosh29/.venv/lib/python3.10/site-packages/torch/cuda/__init__.py:65: FutureWarning: The pynvml package is deprecated. Please install nvidia-ml-py instead. If you did no
t install pynvml directly, please report this to the maintainers of the package that installed pynvml for you.
  import pynvml  # type: ignore[import]

================================================================================
  MSD LIVER TUMOR CLASSIFICATION
  MULTI-MODEL 3-FOLD CROSS-VALIDATION PIPELINE
================================================================================

STEP 1: READING MANIFEST
--------------------------------------------------------------------------------
  Total patches:   5030 (1580 tumor, 3450 non-tumor)
  Unique patients: 118

STEP 2: SETTING UP 3-FOLD CROSS-VALIDATION
--------------------------------------------------------------------------------
  Patient-level splitting (prevents data leakage):

    Fold 1:  39 patients, 1645 patches (527 tumor, 1118 non-tumor)
    Fold 2:  38 patients, 1726 patches (526 tumor, 1200 non-tumor)
    Fold 3:  41 patients, 1659 patches (527 tumor, 1132 non-tumor)

  Device: cuda
  GPU:    NVIDIA A100-SXM4-40GB MIG 3g.20gb
  VRAM:   20.9 GB

  Models to train: ResNet3D, VGG3D
```

```
*******************************************************************************
  TRAINING MODEL 1/2: ResNet3D
  Pipeline elapsed: 0 min
*******************************************************************************

###############################################################################
#  MODEL: ResNet3D (resnet3d)
###############################################################################

===============================================================================
  ResNet3D - FOLD 1 / 3
===============================================================================

  Split: 68 train patients, 11 val patients, 39 test patients
  Datasets:
    train:  2914 patches (895 positive, 2019 negative, ratio 1:2)
    val  :   471 patches (158 positive, 313 negative, ratio 1:1)
    test :  1645 patches (527 positive, 1118 negative, ratio 1:2)
  Parameters: 3.62M (3,615,970 total)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1786 TrA:0.529 | VaL:0.1753 VaA:0.459 VaAUC:0.586 | LR:6.2e-05 | 91s/ep | fold:2m total:2m
    Ep   5/100 | TrL:0.1650 TrA:0.611 | VaL:0.1746 VaA:0.539 VaAUC:0.609 | LR:3.0e-04 | 76s/ep | fold:7m total:7m
    Ep  10/100 | TrL:0.1490 TrA:0.698 | VaL:0.1803 VaA:0.575 VaAUC:0.653 | LR:3.0e-04 | 76s/ep | fold:13m total:13m
    Ep  15/100 | TrL:0.1184 TrA:0.790 | VaL:0.1258 VaA:0.745 VaAUC:0.818 | LR:2.9e-04 | 76s/ep | fold:20m total:20m
    Ep  20/100 | TrL:0.0847 TrA:0.868 | VaL:0.1162 VaA:0.777 VaAUC:0.881 | LR:2.8e-04 | 76s/ep | fold:26m total:26m
    Ep  25/100 | TrL:0.0749 TrA:0.879 | VaL:0.0838 VaA:0.881 VaAUC:0.910 | LR:2.7e-04 | 76s/ep | fold:32m total:32m
    Ep  30/100 | TrL:0.0536 TrA:0.927 | VaL:0.1413 VaA:0.794 VaAUC:0.887 | LR:2.5e-04 | 76s/ep | fold:39m total:39m
    Ep  35/100 | TrL:0.0437 TrA:0.938 | VaL:0.1863 VaA:0.726 VaAUC:0.905 | LR:2.3e-04 | 76s/ep | fold:45m total:45m
    Ep  40/100 | TrL:0.0354 TrA:0.957 | VaL:0.1396 VaA:0.788 VaAUC:0.897 | LR:2.1e-04 | 76s/ep | fold:51m total:51m
    Ep  45/100 | TrL:0.0206 TrA:0.973 | VaL:0.1714 VaA:0.807 VaAUC:0.913 | LR:1.9e-04 | 76s/ep | fold:58m total:58m
    Ep  50/100 | TrL:0.0179 TrA:0.974 | VaL:0.1093 VaA:0.904 VaAUC:0.934 | LR:1.6e-04 | 76s/ep | fold:64m total:64m
    Ep  55/100 | TrL:0.0195 TrA:0.976 | VaL:0.1365 VaA:0.883 VaAUC:0.938 | LR:1.4e-04 | 76s/ep | fold:70m total:70m
    Ep  60/100 | TrL:0.0132 TrA:0.985 | VaL:0.1443 VaA:0.851 VaAUC:0.925 | LR:1.1e-04 | 76s/ep | fold:77m total:77m
```

```
    Ep  65/100 | TrL:0.0103 TrA:0.988 | VaL:0.2817 VaA:0.786 VaAUC:0.926 | LR:9.0e-05 | 76s/ep | fold:83m total:83m
    Ep  70/100 | TrL:0.0079 TrA:0.991 | VaL:0.1586 VaA:0.862 VaAUC:0.929 | LR:6.8e-05 | 76s/ep | fold:89m total:89m
    Ep  75/100 | TrL:0.0048 TrA:0.993 | VaL:0.1721 VaA:0.858 VaAUC:0.928 | LR:4.8e-05 | 76s/ep | fold:96m total:96m
    Ep  80/100 | TrL:0.0042 TrA:0.997 | VaL:0.1798 VaA:0.845 VaAUC:0.932 | LR:3.2e-05 | 77s/ep | fold:102m total:102m
    Ep  85/100 | TrL:0.0028 TrA:0.997 | VaL:0.1859 VaA:0.847 VaAUC:0.935 | LR:1.8e-05 | 76s/ep | fold:108m total:108m
    Ep  90/100 | TrL:0.0015 TrA:0.999 | VaL:0.1722 VaA:0.866 VaAUC:0.934 | LR:8.1e-06 | 76s/ep | fold:115m total:115m
    Ep  95/100 | TrL:0.0016 TrA:0.999 | VaL:0.1759 VaA:0.866 VaAUC:0.934 | LR:2.0e-06 | 76s/ep | fold:121m total:121m
    Ep 100/100 | TrL:0.0017 TrA:0.999 | VaL:0.1772 VaA:0.866 VaAUC:0.933 | LR:0.0e+00 | 76s/ep | fold:127m total:127m

  ResNet3D Fold 1 training complete: 100 epochs in 127.3 min (best val auc: 0.9396, best val loss: 0.1464)
  Estimated time remaining for ResNet3D: ~255 min

  Evaluating ResNet3D fold 1 on test set (1645 samples) with TTA...

============================================================
  EVALUATION METRICS - ResNet3D FOLD 1 TEST
============================================================
  Sensitivity (Recall):  0.8691
  Specificity:           0.8775
  Precision:             0.7697
  Accuracy:              0.8748
  F1 Score:              0.8164
  AUC-ROC:               0.9466

  Confusion Matrix:
    True Positives:   458  (Caught nodules)
    True Negatives:   981  (Correct normals)
    False Positives:  137  (False alarms)
    False Negatives:   69  (Missed nodules)
============================================================
```

```
  Generating Grad-CAM visualizations for ResNet3D...
    TP: 3 figure(s) saved
    TN: 3 figure(s) saved
    FP: 3 figure(s) saved
    FN: 3 figure(s) saved
  Grad-CAM figures saved to: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/gradcam/resnet3d

================================================================================
  ResNet3D - FOLD 2 / 3
================================================================================

  Split: 68 train patients, 12 val patients, 38 test patients
  Datasets:
    train:  2912 patches (864 positive, 2048 negative, ratio 1:2)
    val  :   392 patches (190 positive, 202 negative, ratio 1:1)
    test :  1726 patches (526 positive, 1200 negative, ratio 1:2)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1859 TrA:0.509 | VaL:0.1811 VaA:0.418 VaAUC:0.343 | LR:6.2e-05 | 76s/ep | fold:1m total:131m
    Ep   5/100 | TrL:0.1659 TrA:0.600 | VaL:0.1751 VaA:0.541 VaAUC:0.543 | LR:3.0e-04 | 75s/ep | fold:6m total:136m
    Ep  10/100 | TrL:0.1632 TrA:0.616 | VaL:0.1770 VaA:0.500 VaAUC:0.471 | LR:3.0e-04 | 75s/ep | fold:13m total:142m
    Ep  15/100 | TrL:0.1372 TrA:0.730 | VaL:0.2064 VaA:0.577 VaAUC:0.599 | LR:2.9e-04 | 76s/ep | fold:19m total:148m
    Ep  20/100 | TrL:0.1061 TrA:0.817 | VaL:0.2290 VaA:0.640 VaAUC:0.650 | LR:2.8e-04 | 75s/ep | fold:25m total:155m
    Ep  25/100 | TrL:0.0873 TrA:0.852 | VaL:0.1993 VaA:0.691 VaAUC:0.726 | LR:2.7e-04 | 76s/ep | fold:32m total:161m
    Ep  30/100 | TrL:0.0783 TrA:0.878 | VaL:0.1741 VaA:0.722 VaAUC:0.798 | LR:2.5e-04 | 75s/ep | fold:38m total:167m
    Ep  35/100 | TrL:0.0653 TrA:0.904 | VaL:0.2145 VaA:0.730 VaAUC:0.755 | LR:2.3e-04 | 75s/ep | fold:44m total:173m
    Ep  40/100 | TrL:0.0560 TrA:0.917 | VaL:0.2712 VaA:0.691 VaAUC:0.744 | LR:2.1e-04 | 75s/ep | fold:50m total:180m
    Ep  45/100 | TrL:0.0459 TrA:0.933 | VaL:0.2576 VaA:0.732 VaAUC:0.776 | LR:1.9e-04 | 76s/ep | fold:57m total:186m
    Ep  50/100 | TrL:0.0411 TrA:0.938 | VaL:0.2924 VaA:0.719 VaAUC:0.762 | LR:1.6e-04 | 75s/ep | fold:63m total:192m
    Ep  55/100 | TrL:0.0340 TrA:0.958 | VaL:0.3668 VaA:0.724 VaAUC:0.741 | LR:1.4e-04 | 75s/ep | fold:69m total:199m
    Ep  60/100 | TrL:0.0272 TrA:0.966 | VaL:0.3439 VaA:0.732 VaAUC:0.777 | LR:1.1e-04 | 75s/ep | fold:75m total:205m
```

```
  Early stopping at epoch 61 (no improvement for 25 epochs)

ResNet3D Fold 2 training complete: 61 epochs in 76.7 min (best val auc: 0.8218, best val loss: 0.1928)
Estimated time remaining for ResNet3D: ~102 min

Evaluating ResNet3D fold 2 on test set (1726 samples) with TTA...

============================================================
EVALUATION METRICS - ResNet3D FOLD 2 TEST
============================================================
Sensitivity (Recall):  0.6825
Specificity:           0.9075
Precision:             0.7638
Accuracy:              0.8389
F1 Score:              0.7209
AUC-ROC:               0.8602

Confusion Matrix:
  True Positives:   359  (Caught nodules)
  True Negatives:  1089  (Correct normals)
  False Positives:  111  (False alarms)
  False Negatives:  167  (Missed nodules)
============================================================


================================================================================
ResNet3D - FOLD 3 / 3
================================================================================

Split: 66 train patients, 11 val patients, 41 test patients
Datasets:
  train:  2686 patches (815 positive, 1871 negative, ratio 1:2)
  val  :   685 patches (238 positive, 447 negative, ratio 1:1)
  test :  1659 patches (527 positive, 1132 negative, ratio 1:2)
```

```
  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1756 TrA:0.522 | VaL:0.1709 VaA:0.546 VaAUC:0.648 | LR:6.2e-05 | 73s/ep | fold:1m total:209m
    Ep   5/100 | TrL:0.1670 TrA:0.593 | VaL:0.1637 VaA:0.629 VaAUC:0.684 | LR:3.0e-04 | 74s/ep | fold:6m total:214m
    Ep  10/100 | TrL:0.1607 TrA:0.647 | VaL:0.1665 VaA:0.594 VaAUC:0.708 | LR:3.0e-04 | 76s/ep | fold:12m total:220m
    Ep  15/100 | TrL:0.1589 TrA:0.656 | VaL:0.1703 VaA:0.558 VaAUC:0.694 | LR:2.9e-04 | 73s/ep | fold:19m total:226m
    Ep  20/100 | TrL:0.1542 TrA:0.682 | VaL:0.1621 VaA:0.594 VaAUC:0.633 | LR:2.8e-04 | 73s/ep | fold:25m total:233m
    Ep  25/100 | TrL:0.1514 TrA:0.678 | VaL:0.1802 VaA:0.531 VaAUC:0.648 | LR:2.7e-04 | 75s/ep | fold:31m total:239m
    Ep  30/100 | TrL:0.1325 TrA:0.754 | VaL:0.1776 VaA:0.578 VaAUC:0.669 | LR:2.5e-04 | 72s/ep | fold:37m total:245m
    Ep  35/100 | TrL:0.1199 TrA:0.784 | VaL:0.1619 VaA:0.704 VaAUC:0.728 | LR:2.3e-04 | 72s/ep | fold:43m total:251m
    Ep  40/100 | TrL:0.1087 TrA:0.814 | VaL:0.1833 VaA:0.612 VaAUC:0.709 | LR:2.1e-04 | 72s/ep | fold:49m total:257m
    Ep  45/100 | TrL:0.0889 TrA:0.853 | VaL:0.2101 VaA:0.673 VaAUC:0.733 | LR:1.9e-04 | 72s/ep | fold:55m total:263m
    Ep  50/100 | TrL:0.0861 TrA:0.870 | VaL:0.2212 VaA:0.635 VaAUC:0.714 | LR:1.6e-04 | 72s/ep | fold:61m total:269m
    Ep  55/100 | TrL:0.0694 TrA:0.894 | VaL:0.2290 VaA:0.686 VaAUC:0.773 | LR:1.4e-04 | 72s/ep | fold:67m total:275m
    Ep  60/100 | TrL:0.0587 TrA:0.917 | VaL:0.2884 VaA:0.613 VaAUC:0.737 | LR:1.1e-04 | 72s/ep | fold:73m total:281m
    Ep  65/100 | TrL:0.0461 TrA:0.934 | VaL:0.2836 VaA:0.664 VaAUC:0.745 | LR:9.0e-05 | 73s/ep | fold:79m total:287m
    Ep  70/100 | TrL:0.0406 TrA:0.940 | VaL:0.4511 VaA:0.593 VaAUC:0.733 | LR:6.8e-05 | 72s/ep | fold:85m total:293m
    Ep  75/100 | TrL:0.0314 TrA:0.955 | VaL:0.3722 VaA:0.636 VaAUC:0.733 | LR:4.8e-05 | 72s/ep | fold:91m total:299m
    Ep  80/100 | TrL:0.0281 TrA:0.962 | VaL:0.3685 VaA:0.657 VaAUC:0.739 | LR:3.2e-05 | 72s/ep | fold:97m total:305m

    Early stopping at epoch 80 (no improvement for 25 epochs)

  ResNet3D Fold 3 training complete: 80 epochs in 97.4 min (best val auc: 0.7730, best val loss: 0.2290)

  Evaluating ResNet3D fold 3 on test set (1659 samples) with TTA...

============================================================
  EVALUATION METRICS - ResNet3D FOLD 3 TEST
============================================================
  Sensitivity (Recall):  0.5180
  Specificity:           0.7730
  Precision:             0.5151
  Accuracy:              0.6920
  F1 Score:              0.5166
  AUC-ROC:               0.6823
```

```
  Confusion Matrix:
    True Positives:   273  (Caught nodules)
    True Negatives:   875  (Correct normals)
    False Positives:  257  (False alarms)
    False Negatives:  254  (Missed nodules)
============================================================


================================================================================
  ResNet3D AGGREGATE RESULTS (3-FOLD CV)
================================================================================

  Metric            Fold 1   Fold 2   Fold 3     Mean+/-Std
  -------------------------------------------------------
  sensitivity       0.8691   0.6825   0.5180 0.6899+/-0.1434
  specificity       0.8775   0.9075   0.7730 0.8526+/-0.0577
  precision         0.7697   0.7638   0.5151 0.6829+/-0.1187
  accuracy          0.8748   0.8389   0.6920 0.8019+/-0.0791
  auc_roc           0.9466   0.8602   0.6823 0.8297+/-0.1100
  f1                0.8164   0.7209   0.5166 0.6846+/-0.1251

  Aggregate (all folds combined, 5030 samples):
    sensitivity     0.6899
    specificity     0.8536
    precision       0.6834
    accuracy        0.8022
    auc_roc         0.8450
    f1              0.6866

  ROC curve saved: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/resnet3d/roc_curve_kfold.png
  Training curves saved: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/resnet3d/training_curves_kfold.png
```

```
********************************************************************************
  TRAINING MODEL 2/2: VGG3D
  Pipeline elapsed: 307 min
********************************************************************************

################################################################################
#  MODEL: VGG3D (vgg3d)
################################################################################

================================================================================
  VGG3D - FOLD 1 / 3
================================================================================

  Split: 68 train patients, 11 val patients, 39 test patients
  Datasets:
    train:  2914 patches (895 positive, 2019 negative, ratio 1:2)
    val  :   471 patches (158 positive, 313 negative, ratio 1:1)
    test :  1645 patches (527 positive, 1118 negative, ratio 1:2)
  Parameters: 0.87M (866,338 total)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1882 TrA:0.493 | VaL:0.1774 VaA:0.355 VaAUC:0.583 | LR:6.2e-05 | 50s/ep | fold:1m total:308m
    Ep   5/100 | TrL:0.1653 TrA:0.607 | VaL:0.1704 VaA:0.575 VaAUC:0.633 | LR:3.0e-04 | 49s/ep | fold:4m total:311m
    Ep  10/100 | TrL:0.1505 TrA:0.697 | VaL:0.1621 VaA:0.658 VaAUC:0.757 | LR:3.0e-04 | 48s/ep | fold:8m total:315m
    Ep  15/100 | TrL:0.1334 TrA:0.745 | VaL:0.1718 VaA:0.605 VaAUC:0.713 | LR:2.9e-04 | 48s/ep | fold:12m total:319m
    Ep  20/100 | TrL:0.1216 TrA:0.771 | VaL:0.1535 VaA:0.694 VaAUC:0.813 | LR:2.8e-04 | 48s/ep | fold:16m total:323m
    Ep  25/100 | TrL:0.1060 TrA:0.813 | VaL:0.1763 VaA:0.616 VaAUC:0.787 | LR:2.7e-04 | 48s/ep | fold:20m total:327m
    Ep  30/100 | TrL:0.0948 TrA:0.842 | VaL:0.1336 VaA:0.773 VaAUC:0.816 | LR:2.5e-04 | 48s/ep | fold:24m total:331m
    Ep  35/100 | TrL:0.0828 TrA:0.872 | VaL:0.2304 VaA:0.605 VaAUC:0.836 | LR:2.3e-04 | 48s/ep | fold:28m total:335m
    Ep  40/100 | TrL:0.0728 TrA:0.888 | VaL:0.1814 VaA:0.690 VaAUC:0.810 | LR:2.1e-04 | 48s/ep | fold:32m total:339m
    Ep  45/100 | TrL:0.0619 TrA:0.912 | VaL:0.2485 VaA:0.648 VaAUC:0.820 | LR:1.9e-04 | 48s/ep | fold:36m total:344m
```

```
    Ep  50/100 | TrL:0.0505 TrA:0.927 | VaL:0.2060 VaA:0.650 VaAUC:0.866 | LR:1.6e-04 | 48s/ep | fold:40m total:348m
    Ep  55/100 | TrL:0.0407 TrA:0.944 | VaL:0.1757 VaA:0.703 VaAUC:0.830 | LR:1.4e-04 | 48s/ep | fold:44m total:352m
    Ep  60/100 | TrL:0.0368 TrA:0.948 | VaL:0.1709 VaA:0.756 VaAUC:0.842 | LR:1.1e-04 | 48s/ep | fold:48m total:356m
    Ep  65/100 | TrL:0.0267 TrA:0.962 | VaL:0.2218 VaA:0.711 VaAUC:0.851 | LR:9.0e-05 | 48s/ep | fold:52m total:360m
    Ep  70/100 | TrL:0.0196 TrA:0.976 | VaL:0.2988 VaA:0.658 VaAUC:0.796 | LR:6.8e-05 | 48s/ep | fold:57m total:364m
    Ep  75/100 | TrL:0.0171 TrA:0.981 | VaL:0.2496 VaA:0.720 VaAUC:0.830 | LR:4.8e-05 | 49s/ep | fold:61m total:368m
    Ep  80/100 | TrL:0.0150 TrA:0.983 | VaL:0.1994 VaA:0.779 VaAUC:0.872 | LR:3.2e-05 | 48s/ep | fold:65m total:372m

    Early stopping at epoch 83 (no improvement for 25 epochs)

  VGG3D Fold 1 training complete: 83 epochs in 67.0 min (best val auc: 0.8885, best val loss: 0.1275)
  Estimated time remaining for VGG3D: ~134 min

  Evaluating VGG3D fold 1 on test set (1645 samples) with TTA...

============================================================
  EVALUATION METRICS - VGG3D FOLD 1 TEST
============================================================
  Sensitivity (Recall):  0.8140
  Specificity:           0.9338
  Precision:             0.8529
  Accuracy:              0.8954
  F1 Score:              0.8330
  AUC-ROC:               0.9325

  Confusion Matrix:
    True Positives:   429  (Caught nodules)
    True Negatives:  1044  (Correct normals)
    False Positives:   74  (False alarms)
    False Negatives:   98  (Missed nodules)
============================================================


  Generating Grad-CAM visualizations for VGG3D...
    TP: 3 figure(s) saved
```

```
    FP: 3 figure(s) saved
    FN: 3 figure(s) saved
  Grad-CAM figures saved to: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/gradcam/vgg3d

================================================================================
  VGG3D - FOLD 2 / 3
================================================================================

  Split: 68 train patients, 12 val patients, 38 test patients
  Datasets:
    train:  2912 patches (864 positive, 2048 negative, ratio 1:2)
    val  :   392 patches (190 positive, 202 negative, ratio 1:1)
    test :  1726 patches (526 positive, 1200 negative, ratio 1:2)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1918 TrA:0.510 | VaL:0.1717 VaA:0.577 VaAUC:0.552 | LR:6.2e-05 | 49s/ep | fold:1m total:377m
    Ep   5/100 | TrL:0.1590 TrA:0.651 | VaL:0.1796 VaA:0.564 VaAUC:0.551 | LR:3.0e-04 | 48s/ep | fold:4m total:380m
    Ep  10/100 | TrL:0.0859 TrA:0.866 | VaL:0.1701 VaA:0.699 VaAUC:0.731 | LR:3.0e-04 | 48s/ep | fold:8m total:384m
    Ep  15/100 | TrL:0.0703 TrA:0.897 | VaL:0.1753 VaA:0.747 VaAUC:0.799 | LR:2.9e-04 | 48s/ep | fold:12m total:388m
    Ep  20/100 | TrL:0.0525 TrA:0.927 | VaL:0.2155 VaA:0.732 VaAUC:0.805 | LR:2.8e-04 | 48s/ep | fold:16m total:392m
    Ep  25/100 | TrL:0.0460 TrA:0.939 | VaL:0.2065 VaA:0.765 VaAUC:0.862 | LR:2.7e-04 | 48s/ep | fold:20m total:396m
    Ep  30/100 | TrL:0.0342 TrA:0.955 | VaL:0.1928 VaA:0.796 VaAUC:0.876 | LR:2.5e-04 | 48s/ep | fold:24m total:400m
    Ep  35/100 | TrL:0.0283 TrA:0.958 | VaL:0.2541 VaA:0.781 VaAUC:0.853 | LR:2.3e-04 | 48s/ep | fold:28m total:404m
    Ep  40/100 | TrL:0.0271 TrA:0.964 | VaL:0.2326 VaA:0.804 VaAUC:0.888 | LR:2.1e-04 | 48s/ep | fold:32m total:408m
    Ep  45/100 | TrL:0.0222 TrA:0.970 | VaL:0.2777 VaA:0.781 VaAUC:0.848 | LR:1.9e-04 | 48s/ep | fold:36m total:412m
    Ep  50/100 | TrL:0.0213 TrA:0.975 | VaL:0.2666 VaA:0.801 VaAUC:0.878 | LR:1.6e-04 | 48s/ep | fold:40m total:416m
    Ep  55/100 | TrL:0.0141 TrA:0.985 | VaL:0.3744 VaA:0.786 VaAUC:0.819 | LR:1.4e-04 | 48s/ep | fold:44m total:420m

    Early stopping at epoch 58 (no improvement for 25 epochs)

  VGG3D Fold 2 training complete: 58 epochs in 46.3 min (best val auc: 0.9116, best val loss: 0.1619)
  Estimated time remaining for VGG3D: ~57 min
```

```
============================================================
  EVALUATION METRICS - VGG3D FOLD 2 TEST
============================================================
  Sensitivity (Recall):  0.8688
  Specificity:           0.9558
  Precision:             0.8961
  Accuracy:              0.9293
  F1 Score:              0.8822
  AUC-ROC:               0.9577

  Confusion Matrix:
    True Positives:   457  (Caught nodules)
    True Negatives:  1147  (Correct normals)
    False Positives:   53  (False alarms)
    False Negatives:   69  (Missed nodules)
============================================================


================================================================================
  VGG3D - FOLD 3 / 3
================================================================================

  Split: 66 train patients, 11 val patients, 41 test patients
  Datasets:
    train:  2686 patches (815 positive, 1871 negative, ratio 1:2)
    val  :   685 patches (238 positive, 447 negative, ratio 1:1)
    test :  1659 patches (527 positive, 1132 negative, ratio 1:2)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1874 TrA:0.498 | VaL:0.1751 VaA:0.467 VaAUC:0.646 | LR:6.2e-05 | 46s/ep | fold:1m total:424m
    Ep   5/100 | TrL:0.1656 TrA:0.615 | VaL:0.1663 VaA:0.599 VaAUC:0.696 | LR:3.0e-04 | 46s/ep | fold:4m total:427m
    Ep  10/100 | TrL:0.1514 TrA:0.689 | VaL:0.1508 VaA:0.653 VaAUC:0.767 | LR:3.0e-04 | 46s/ep | fold:8m total:431m
    Ep  15/100 | TrL:0.1316 TrA:0.758 | VaL:0.1384 VaA:0.676 VaAUC:0.823 | LR:2.9e-04 | 46s/ep | fold:11m total:435m
```

```
  Ep  20/100 | TrL:0.1060 TrA:0.829 | VaL:0.1008 VaA:0.816 VaAUC:0.920 | LR:2.8e-04 | 46s/ep | fold:15m total:439m
  Ep  25/100 | TrL:0.0845 TrA:0.863 | VaL:0.0621 VaA:0.940 VaAUC:0.967 | LR:2.7e-04 | 46s/ep | fold:19m total:443m
  Ep  30/100 | TrL:0.0664 TrA:0.899 | VaL:0.0600 VaA:0.915 VaAUC:0.984 | LR:2.5e-04 | 46s/ep | fold:23m total:446m
  Ep  35/100 | TrL:0.0558 TrA:0.918 | VaL:0.0758 VaA:0.845 VaAUC:0.978 | LR:2.3e-04 | 46s/ep | fold:27m total:450m
  Ep  40/100 | TrL:0.0472 TrA:0.932 | VaL:0.0334 VaA:0.966 VaAUC:0.985 | LR:2.1e-04 | 46s/ep | fold:31m total:454m
  Ep  45/100 | TrL:0.0408 TrA:0.947 | VaL:0.0348 VaA:0.966 VaAUC:0.989 | LR:1.9e-04 | 46s/ep | fold:34m total:458m
  Ep  50/100 | TrL:0.0254 TrA:0.965 | VaL:0.0647 VaA:0.909 VaAUC:0.987 | LR:1.6e-04 | 46s/ep | fold:38m total:462m
  Ep  55/100 | TrL:0.0220 TrA:0.971 | VaL:0.0334 VaA:0.964 VaAUC:0.989 | LR:1.4e-04 | 46s/ep | fold:42m total:465m
  Ep  60/100 | TrL:0.0175 TrA:0.977 | VaL:0.0280 VaA:0.977 VaAUC:0.991 | LR:1.1e-04 | 46s/ep | fold:46m total:469m
  Ep  65/100 | TrL:0.0180 TrA:0.976 | VaL:0.0240 VaA:0.971 VaAUC:0.991 | LR:9.0e-05 | 46s/ep | fold:50m total:473m
  Ep  70/100 | TrL:0.0100 TrA:0.990 | VaL:0.0627 VaA:0.920 VaAUC:0.987 | LR:6.8e-05 | 46s/ep | fold:53m total:477m
  Ep  75/100 | TrL:0.0108 TrA:0.989 | VaL:0.0317 VaA:0.961 VaAUC:0.990 | LR:4.8e-05 | 46s/ep | fold:57m total:481m
  Ep  80/100 | TrL:0.0059 TrA:0.993 | VaL:0.0488 VaA:0.945 VaAUC:0.989 | LR:3.2e-05 | 46s/ep | fold:61m total:485m
  Ep  85/100 | TrL:0.0053 TrA:0.995 | VaL:0.0328 VaA:0.968 VaAUC:0.989 | LR:1.8e-05 | 46s/ep | fold:65m total:488m
  Ep  90/100 | TrL:0.0068 TrA:0.994 | VaL:0.0355 VaA:0.968 VaAUC:0.989 | LR:8.1e-06 | 46s/ep | fold:69m total:492m
  Ep  95/100 | TrL:0.0043 TrA:0.996 | VaL:0.0327 VaA:0.971 VaAUC:0.990 | LR:2.0e-06 | 46s/ep | fold:73m total:496m
  Ep 100/100 | TrL:0.0028 TrA:0.998 | VaL:0.0328 VaA:0.972 VaAUC:0.990 | LR:0.0e+00 | 46s/ep | fold:76m total:500m

VGG3D Fold 3 training complete: 100 epochs in 76.5 min (best val auc: 0.9920, best val loss: 0.0329)

Evaluating VGG3D fold 3 on test set (1659 samples) with TTA...

============================================================
EVALUATION METRICS - VGG3D FOLD 3 TEST
============================================================
Sensitivity (Recall):  0.8539
Specificity:           0.9435
Precision:             0.8755
Accuracy:              0.9150
F1 Score:              0.8646
AUC-ROC:               0.9490

Confusion Matrix:
  True Positives:   450  (Caught nodules)
  True Negatives:  1068  (Correct normals)
```

```
    False Positives:   64  (False alarms)
    False Negatives:   77  (Missed nodules)
============================================================


================================================================================
  VGG3D AGGREGATE RESULTS (3-FOLD CV)
================================================================================

  Metric            Fold 1   Fold 2   Fold 3     Mean+/-Std
  -------------------------------------------------------
  sensitivity       0.8140   0.8688   0.8539 0.8456+/-0.0231
  specificity       0.9338   0.9558   0.9435 0.9444+/-0.0090
  precision         0.8529   0.8961   0.8755 0.8748+/-0.0176
  accuracy          0.8954   0.9293   0.9150 0.9133+/-0.0139
  auc_roc           0.9325   0.9577   0.9490 0.9464+/-0.0105
  f1                0.8330   0.8822   0.8646 0.8599+/-0.0204

  Aggregate (all folds combined, 5030 samples):
    sensitivity     0.8456
    specificity     0.9446
    precision       0.8749
    accuracy        0.9135
    auc_roc         0.9429
    f1              0.8600

  ROC curve saved: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/vgg3d/roc_curve_kfold.png
  Training curves saved: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/vgg3d/training_curves_kfold.png
```

```
================================================================================
  STEP 4: CROSS-MODEL COMPARISON
================================================================================

  Model           Params     sensitivity     specificity     precision       accuracy        auc_roc         f1              Time
  ----------------------------------------------------------------------------------------------------------------------------------
  ResNet3D        3.62M      0.690+/-0.143  0.853+/-0.058  0.683+/-0.119  0.802+/-0.079  0.830+/-0.110  0.685+/-0.125  307m
  VGG3D           0.87M      0.846+/-0.023  0.944+/-0.009  0.875+/-0.018  0.913+/-0.014  0.946+/-0.010  0.860+/-0.020  194m

  Comparison ROC saved: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/comparison_roc.png
  Results summary saved: /nlsasfs/home/aikosh/prod-aikosh29/run_liver/liver/results/liver/plots/results_summary.txt
```